

PART G — WEAPON CARDS (the last-day sheet; Sunday night + exam-morning read)

One card = one reflex. If you can recall all 30 of these, you can execute the whole exam.

Arithmetic weapons

1. **Fraction ladder** — $1/8 = 12.5\%$, $1/6 = 16.66\%$, $1/12 = 8.33\%$, $1/7 = 14.28\%$, $1/16 = 6.25\%$, $1/3 = 33.33\%$. Any % question is a fraction question.
2. **Change of base** — "A is $r\%$ more than B; B is less by?" $\rightarrow r/(100+r) \times 100$. (25% more $\rightarrow$ 20% less.)
3. **Successive changes** — a then $b \rightarrow a + b + ab/100$. Discounts multiply DOWN: $\times 4/5 \times 9/10 = 18/25$.
4. **Salary equation** — $\text{rent}\% + \text{fixed spends} + \text{savings} = \text{salary} \rightarrow \text{fixed} + \text{savings} = (100-r)\%$ of salary. One division.
5. **Growth chains** — $\times 11/10$, $\times 6/5$, $\times 9/10$ multiply as fractions; convert to a single fraction and cancel.
6. **CI ladder** — $10\% \rightarrow 21\% \rightarrow 33.1\%$; $20\% \rightarrow 44\% \rightarrow 72.8\%$. $\text{CI-SI (2 yr)} = P(r/100)^2$.
7. **Equal profit%/loss%** — $\text{CP} = (\text{SP}_1 + \text{SP}_2)/2$.
8. **SP of n = CP of m** — $\text{gain}\% = (m-n)/n \times 100$ with sign discipline.
9. **HCF $\times$ LCM = product** (two numbers only). $\text{HCF} = k$ for $2k, 3k, 4k \rightarrow \text{LCM} = 12k$.
10. **Sum+product $\rightarrow$ difference** — $\sqrt{S^2 - 4P}$ where S = sum, P = product (difference of roots = $\sqrt{((x+y)^2 - 4xy)}$). ($S=76$, $P=1408 \rightarrow \sqrt{5776-5632} = 12$.)
11. **LCM-units for work** — total = LCM of times; rate = LCM/time; add for together, subtract for empty.
12. **Alternate-day work** — 2-day cycle = $a+b$ units; count cycles, then leftover $\div$ next day's rate.
13. **Average shortcut** — pairwise averages: $a+b$, $b+c$, $a+c$ over 2. Weighted average pulls toward the bigger weight.
14. **Alligation cross** — $(d-m):(m-c)$ gives the mixing ratio. Replacement: $\text{final} = \text{init} \times (1-x/V)^n$.
15. **Average speed** — unequal legs: total distance $\div$ total time; equal legs: $2ab/(a+b)$. NOT the arithmetic mean.
16. **Trains** — pole: length = speed $\times$ time; train+platform: length = (sum of lengths); opposite: speeds add, same: subtract. $\text{km/h} \rightarrow \text{m/s} \times 5/18$.
17. **Meet-up with head start** — subtract the head-start distance from the gap, then close at relative speed.
18. **Circular track** — same direction: $C/\Delta v$; opposite: $C/\Sigma v$; together-at-start: LCM of lap times.
19. **Divisibility-11 with digits** — (sum odd positions) - (sum even positions) = 0 or ± 11 .
20. **Divisible by 24/36/72** — split into coprime factors and satisfy BOTH ($24 = 3 \times 8$).
21. **Unit digits** — cycles of 4 for 2,3,7,8; constant for 0,1,5,6; 2-cycle for 4,9. Exponent mod cycle.
22. **Recurring decimal** — $0.\text{abab}\dots = \text{ab}/99$; $0.\text{abcabc}\dots = \text{abc}/999$.

Geometry / mensuration / stats weapons

1. **Cylinder from circumference** — $C = 2\pi r \rightarrow r = C/2\pi$; capacity $L = \pi r^2 h/1000$. This exact template has appeared in every shift.
2. **Melt-recast invariant** — volume conserved: count = $(R/r)^3$; then $\text{TSA} = \text{count} \times 4\pi r^2$.
3. **Chord** — half-chord = $\sqrt{r^2 - d^2}$. Pythagorean triples save you: 3-4-5, 5-12-13, 7-24-25, 9-40-41, 12-35-37.
4. **Isosceles area** — height = $\sqrt{a^2 - (\text{base}/2)^2}$; equal sides 41, base 18 $\rightarrow$ 9-40-41 $\rightarrow$ area 360.
5. **Tangent-secant** — $PT^2 = PA \times PB$. Angle in semicircle = 90° . Same-base triangles: area ratio = height ratio.
6. **Empirical relation** — mode = $3 \cdot \text{median} - 2 \cdot \text{mean}$. AP-spaced data $\text{SD} = d \cdot \sqrt{(n^2 - 1)/12}$.

Reasoning weapons

1. **Syllogism core** — $A+A \rightarrow A$; $A+I \rightarrow I$; reverse of All = Some; "Some-not" gives nothing definite; either-or needs a complementary pair. Possibility $\neq$ definite conclusion.
2. **Clock single formula** — $|30H - 5.5M|$. Together: $60H/11$ min past H. Mirror: $11:60 - t$.
3. **Calendar odd days** — $100y=5$, $200y=3$, $300y=1$, $400y=0$. Ordinary+6 or +11, leap+28.
4. **Ranking** — total = top + bottom - 1 (overlap) or +gap; between = $|\text{positions}| - 1$.
5. **Series ladder** — $d_1 \rightarrow d_2 \rightarrow d_3$; then ratio; then $n^2 \pm k$, $n^3 \pm k$; alternating twins last. (91,92,96,97,101 $\rightarrow$ +1,+4 alternating $\rightarrow$ 102.)

6. **Coding 20-second protocol** — compute the shift on 2 letters; consistent → extrapolate; not a shift → try positions, reversal, vowel/consonant maps.
7. **RC 90-second protocol** — read questions first → mark structure words (But/However/Therefore) and tone words → answer the 5 archetypes from the map: main idea = the author's POINT, application = same lesson new situation, structure = how it's organised, tone = from adjectives.
8. **Number-symbol series** — write target POSITIONS first, then apply left/right and odd/even constraints. Never count live.
9. **Guessing gate** — only on 50/50 ($EV + \frac{1}{3}$). Blind 4-way guess $EV = 0$ and burns 1.33 when wrong.

Exam-morning mini-recital (60 seconds)

Fraction ladder → squares 21–29 → the 4 cylinder-template steps → clock formula → syllogism table → "guess only 50/50".